

Proposition 5.1

parabolic rescaling		Han--Li--Sun		Proposition 5.1		(X_0,T)	
Andrews--Chow--Guenther--Langford		Extrinsic Geometric Flows		369			
$X_j(x,t)=\lambda_j\Big(X(x,t_j+\lambda_j^{-2}t)-X(x_j,t_j)\Big),\qquad \lambda_j=H(x_j,t_j).$							
Proposition 5.1		spacetime L^2					
mean curvature flow		shrinking sphere		$ H_\infty (0,0)=1$		smooth pointed	
blow-up limit							
Han--Li--Sun		beta-symplectic gradient flow		fixed-			
center		Proposition 5.1					
$q_j=\lambda_j\big(F(x_j,t_j)-X_0\big),\qquad \theta_j=\lambda_j^2(T-t_j).$							
q_j	θ_j	H_j,V_j,f_j	L^2	$F_j^\perp$	$q_j^\perp$	L^2	
$q_j\rightarrow 0$		spacetime area bound		Han--Li--Sun		moving-cutoff	
(5.5)							
				$F(x_j,t_j)\rightarrow X_0$		$q_j\rightarrow 0$	
$ F(x_j,t_j)-X_0 =o(\lambda_j^{-1});$							
essential Type-I		$\lambda_j\asymp (T-t_j)^{-1/2}$		$o(\sqrt{T-t_j})$		$\lambda_j= H , A ,(T-$	
$t_j)^{-1/2}$		q_j		$q_j\rightarrow 0$		Chen--Li	
4.7		q_j		x_k		p	
		$q=0$		cutoff			

1.

$F : \Sigma \times [0, T) \rightarrow M$	beta-symplectic critical-surface gradient flow	normal coordinates
$\partial_\tau F = f, \quad f = \frac{\cos^2 \alpha H - \beta \sin^2 \alpha V}{\cos^2 \alpha + \beta \sin^2 \alpha},$		

H mean-curvature vector V Han--Li--Sun normal field α Kähler angle
 symplectic-angle bound

$$\cos \alpha \geq \delta > 0.$$

Han--Li--Sun (X_0,T)

$$G_\lambda(x,u)=\lambda\big(F(x,T+\lambda^{-2}u)-X_0\big),\qquad u<0.$$

Proposition 5.1 $R>0$ $-\infty < a < b < 0$ $\lambda \rightarrow \infty$

$$\begin{aligned} \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |H_\lambda|^2 \, d\mu_u^\lambda \, du, \quad & \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |V_\lambda|^2 \, d\mu_u^\lambda \, du, \\ \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |f_\lambda|^2 \, d\mu_u^\lambda \, du, \quad & \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |G_\lambda^\perp|^2 \, d\mu_u^\lambda \, du. \end{aligned}$$

369 $x_j \in \Sigma \quad t_j \nearrow T$

$$p_j = F(x_j,t_j), \qquad \lambda_j = |H|(x_j,t_j) \longrightarrow \infty,$$

$$F_j(x,t)=\lambda_j\Big(F(x,t_j+\lambda_j^{-2}t)-p_j\Big). \tag{1.1}$$

rescaled time interval

$$F_j(x_j,0)=0, \qquad |H_j|(x_j,0)=1. \tag{1.2}$$

(1.1) $R>0$ $s_1 < s_2 < 0$ Proposition 5.1

2. scaling identities

$$\tau_j(t)=t_j+\lambda_j^{-2}t.$$

translation tangent normal spaces ambient dilation $y \mapsto \lambda_j y$ induced metric λ_j^2 Σ

$$g_j=\lambda_j^2g, \qquad d\mu_t^j=\lambda_j^2d\mu_{\tau_j(t)}, \qquad d\tau=\lambda_j^{-2}dt. \tag{2.1}$$

second fundamental form inverse metric

$$H_j=\lambda_j^{-1}H. \tag{2.2}$$

Kähler angle dilation scaled orthonormal tangent frame λ_j^{-1}

$$V_j=\lambda_j^{-1}V. \tag{2.3}$$

$$f_j = \frac{\cos^2 \alpha_j H_j - \beta \sin^2 \alpha_j V_j}{\cos^2 \alpha_j + \beta \sin^2 \alpha_j} = \lambda_j^{-1} f, \quad \partial_t F_j = f_j. \quad (2.4)$$

$\cos \alpha_j \geq \delta$ uniform pointwise estimate

$$|f_j|^2 \leq C(\beta, \delta) (|H_j|^2 + |V_j|^2). \quad (2.5)$$

fixed singular point (X_0, T)

$$q_j = \lambda_j(p_j - X_0), \quad \theta_j = \lambda_j^2(T - t_j). \quad (2.6)$$

fixed-center coordinates

$$G_j(x, t) := \lambda_j(F(x, \tau_j(t)) - X_0) = F_j(x, t) + q_j, \quad (2.7)$$

$$T - \tau_j(t) = \lambda_j^{-2}(\theta_j - t). \quad (2.8)$$

backward heat kernel

$$\rho_{X_0, T}(F, \tau_j(t)) d\mu_{\tau_j(t)} = \psi_j(F_j, t) d\mu_t^j, \quad (2.9)$$

$$\psi_j(z, t) = \frac{1}{4\pi(\theta_j - t)} \exp\left(-\frac{|z + q_j|^2}{4(\theta_j - t)}\right). \quad (2.10)$$

normal-position identity

$$G_j^\perp = (F_j + q_j)^\perp = F_j^\perp + q_j^\perp. \quad (2.11)$$

fixed-center monotonicity

$$(F_j + q_j)^\perp \quad F_j^\perp$$

3. Gaussian lower bound sharp criterion

$R > 0$ $s_1 < s_2 < 0$ cylinder

$$|F_j| \leq R, \quad s_1 \leq t \leq s_2$$

j $c_R > 0$ $\psi_j \geq c_R$

$$\sup_j |q_j| < \infty, \quad \sup_j \theta_j < \infty. \quad (3.1)$$

$$|q_j| \leq Q \quad 0 \leq \theta_j \leq \Theta$$

$$-s_2 \leq \theta_j - t \leq \Theta - s_1, \quad |F_j + q_j| \leq R + Q,$$

$$\psi_j(F_j,t)\geq \frac{1}{4\pi(\Theta-s_1)}\exp\left(-\frac{(R+Q)^2}{4(-s_2)}\right)>0. \tag{3.2}$$

$$\begin{array}{ccccccc} \theta_j \rightarrow \infty & z=0, t=s_2 & \text{prefactor} & |q_j| \rightarrow \infty & \theta_j & \text{exponential factor} \\ \text{ball} & (3.1) & \text{uniform Gaussian lower bound} & & & \end{array}$$

4. “ ”

4.1 shrinking sphere

$$\begin{array}{ccccccc} \text{ordinary mean curvature flow} & \text{beta-symplectic-specific counterexample} & & & & & \text{“} \quad 369 \\ \text{curvature-normalizing parametrization} & & & & & & \text{”} \end{array}$$

$$M_t=S^n_{r(t)}\subset \mathbb{R}^{n+1} \quad \text{shrinking round sphere}$$

$$r(t)=\sqrt{2n(T-t)}.$$

$$e \quad t_j \nearrow T \quad p_j=r(t_j)e$$

$$\lambda_j=|H|(p_j,t_j)=\frac{n}{r(t_j)}.$$

$$(1.1) \qquad j \qquad \text{rescaled flow} \qquad \text{sphere center} \quad -ne \quad \text{radius}$$

$$r_j(t)=\sqrt{n^2-2nt},$$

$$|H_j|(t)=\frac{n}{\sqrt{n^2-2nt}}, \qquad |H_j|(x_j,0)=1. \tag{4.1}$$

$$\varepsilon>0 \qquad s_1=-\varepsilon \quad s_2=-\varepsilon/2$$

$$R>\sqrt{n^2+2n\varepsilon}-n,$$

$$t\in[s_1,s_2] \quad \text{sphere} \quad B_R(0) \qquad \text{spherical cap}$$

$$\int_{s_1}^{s_2}\int_{M_t^j\cap B_R(0)}|H_j|^2\,d\mu_t^j\,dt$$

$$j \qquad \text{Sun--Jun} \qquad \text{surface dimension} \qquad n=2$$

4.2 smooth nonflat limit obstruction

$$\begin{array}{ccc} \text{round symmetry} & \text{moving-center rescalings} & \text{smooth locally converges} \\ F_\infty & \text{normalization} & \end{array}$$

$$|H_\infty|(p_\infty,0)=1.$$

$$\text{compact coordinate patch } K \quad s_1 < s_2 < 0$$

$$|H_\infty| \geq \frac{1}{2} \quad \text{on } K \times [s_1, s_2].$$

smooth convergence j $|H_j| \geq 1/4$ induced area density patch R
 $F_j(K, t) \subset B_R(0)$ $c > 0$

$$\liminf_{j \rightarrow \infty} \int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 d\mu_t^j dt \geq c > 0. \quad (4.2)$$

beta-symplectic curvature-normalizing sequence smooth nonflat blow-up limit
 $|H_j|(x_j, 0) = 1$ compactness (4.2)

regularity

5. beta-symplectic flow

Han--Li--Sun fixed-center Proposition 5.1

Theorem 5.1 moving-center transfer

F beta-symplectic gradient flow (X_0, T) fixed singular point $\lambda_j \rightarrow \infty$ (1.1) (2.6)
 F_j, q_j, θ_j

1. λ_j fixed-center rescalings

$$\tilde{F}_j(x, u) = \lambda_j (F(x, T + \lambda_j^{-2} u) - X_0)$$

Han--Li--Sun Proposition 5.1 fixed-center spacetime L^2

2. $\sup_j |q_j| < \infty$ $0 \leq \theta_j \leq \Theta < \infty$

3. $[s_1, s_2] \subset (-\infty, 0)$ j moving rescaled time domain

$R > 0$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.1)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |V_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.2)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |f_j|^2 d\mu_t^j dt \rightarrow 0. \quad (5.3)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |F_j^\perp|^2 d\mu_t^j dt \rightarrow 0, \quad (5.4)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |q_j^\perp|^2 d\mu_t^j dt \rightarrow 0. \quad (5.5)$$

$$(5.5) \qquad \text{normal-displacement condition} \qquad \text{Han--Li--Sun} \qquad \text{moving-cutoff} \qquad (5.5)$$

$$q_j \rightarrow 0 \qquad (5.6)$$

$$A_R < \infty$$

$$\int_{s_1}^{s_2} \mu_t^j(\Sigma_t^j \cap B_R(0)) \, dt \leq A_R, \qquad (5.7)$$

$$(5.4) \qquad \text{moving-center}$$

Proof

moving time t

$$u = t - \theta_j.$$

$$\theta_j = \lambda_j^2(T - t_j)$$

$$T + \lambda_j^{-2}u = t_j + \lambda_j^{-2}t.$$

$$\widetilde{F}_j(x,u) = F_j(x,t) + q_j. \qquad (5.8)$$

$$\text{translation} \qquad H_j, V_j, f_j \quad \text{normal spaces} \quad \text{induced measure} \quad |F_j(x,t)| < R \quad |q_j| \leq Q$$

$$|\widetilde{F}_j(x,u)| < R + Q.$$

$$0 \leq \theta_j \leq \Theta \qquad t \in [s_1, s_2]$$

$$u \in [s_1 - \Theta, s_2] \subset (-\infty, 0).$$

integrands

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |H_j|^2 \leq \int_{s_1 - \Theta}^{s_2} \int_{\{|\widetilde{F}_j| < R + Q\}} |\widetilde{H}_j|^2.$$

$$\text{fixed-center Proposition 5.1} \qquad (5.1) \qquad \text{inclusion} \quad \text{time shift} \qquad (5.2) \quad (5.3)$$

$$(2.5) \quad (5.1) \quad (5.2) \qquad (5.3)$$

$$\text{normal-position term} \qquad (2.11)$$

$$\widetilde{F}_j^\perp = F_j^\perp + q_j^\perp.$$

$$\text{fixed-center Proposition 5.1} \qquad \text{ball/time inclusion}$$

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |F_j^\perp + q_j^\perp|^2 \rightarrow 0. \qquad (5.9)$$

$$|F_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|q_j^\perp|^2$$

$$(5.5) \Rightarrow (5.4)$$

$$|q_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|F_j^\perp|^2$$

$$(5.9) \qquad (5.4) \Rightarrow (5.5) \qquad \text{orthogonal projection}$$

$$\int |q_j^\perp|^2 \leq |q_j|^2 \int d\mu_t^j dt.$$

$$(5.6) \quad (5.7)$$

6.

$q_j \quad \theta_j$	shifted normal-position limit	$F_j^\perp$ limit	affine plane
$F_j(u,v,t)=(u,v,-1)\subset \mathbb{R}^3, \qquad q_j=(0,0,1).$			
$H_j=V_j=f_j=0$			
$F_j+q_j=(u,v,0)$			
tangent	$(F_j+q_j)^\perp=0$		
$F_j^\perp=(0,0,-1).$			
$R>1 \quad s_1<s_2$			
$\int_{s_1}^{s_2}\int_{F_j\cap B_R(0)} F_j^\perp ^2\,d\mu\,dt=(s_2-s_1)\pi(R^2-1)>0.$			
center displacement	normal component	“	”

$$7. \qquad q_j \rightarrow 0$$

X_0	normal coordinates
$d_j = F(x_j,t_j) - X_0 , \qquad q_j = \lambda_j(F(x_j,t_j) - X_0).$	

Proposition 7.1

$$t_j \nearrow T \quad d_j \rightarrow 0 \quad \lambda_j > 0$$

$$q_j \rightarrow 0 \quad \Longleftrightarrow \quad d_j = o(\lambda_j^{-1}). \tag{7.2}$$

$$|A|^2(x,t) \leq \frac{K}{T-t}, \tag{7.3}$$

$$\begin{aligned} \lambda_j &= |A|(x_j,t_j) \quad \lambda_j = |H|(x_j,t_j) \\ d_j &= o(\sqrt{T-t_j}) \end{aligned} \tag{7.4}$$

$$\begin{aligned} q_j \rightarrow 0 \qquad \text{essential Type-I} \qquad c > 0 \\ \lambda_j \sqrt{T-t_j} \geq c, \\ (7.4) \qquad \lambda_j = (T-t_j)^{-1/2} \quad (7.4) \quad q_j \rightarrow 0 \end{aligned} \tag{7.5}$$

Proof

$$\begin{aligned} |q_j| &= \lambda_j d_j. \\ (7.2) \qquad \text{Cauchy--Schwarz} \end{aligned} \tag{7.6}$$

$$\begin{aligned} |H|^2 &\leq 2|A|^2. \\ (7.3) \quad \lambda_j &= |A| \quad \lambda_j \sqrt{T-t_j} \leq \sqrt{K} \quad \lambda_j = |H| \quad \lambda_j \sqrt{T-t_j} \leq \sqrt{2K} \end{aligned} \tag{7.6}$$

$$\begin{aligned} |q_j| &= \big(\lambda_j \sqrt{T-t_j}\big) \frac{d_j}{\sqrt{T-t_j}}, \\ (7.5) \qquad |q_j| &= d_j/\sqrt{T-t_j} \end{aligned}$$

7.2

$$\begin{aligned} o \qquad d_j \rightarrow 0 \qquad T &= 1 \\ F : S^2 \times [0,1) &\longrightarrow \mathbb{R}^4, \qquad F(u,t) = 2\sqrt{1-t} \, u, \end{aligned} \tag{7.8}$$

$$\begin{aligned} S^2 \qquad \mathbb{R}^4 \qquad \partial_t F &= H \\ |A|^2 &= \frac{1}{2(1-t)}, \qquad |H| = \frac{1}{\sqrt{1-t}}. \end{aligned} \tag{7.9}$$

$$t=1 \qquad u_0 \in S^2 \qquad x_j = u_0 \quad t_j \nearrow 1 \quad X_0 = 0 \qquad x_j$$

$$F(x_j,t_j) = 2\sqrt{1-t_j} \, u_0 \longrightarrow 0.$$

$$\begin{array}{c|c} \lambda_j & q_j = \lambda_j F(x_j,t_j) \\ \hline |H|(x_j,t_j) & 2u_0 \\ |A|(x_j,t_j) & \sqrt{2} \, u_0 \\ (1-t_j)^{-1/2} & 2u_0 \end{array} \tag{7.10}$$

	mean curvature flow	Han--Li--Sun	beta-symplectic flow
“	$q_j \rightarrow 0$		beta-symplectic
	(7.2)	o	

7.3

$q_j \rightarrow 0$	(7.4)	$d_j = o(\lambda_j^{-1})$
$q_j \rightarrow 0$ Chen--Li	q_j	$q_j \rightarrow q$

8. Chen--Li 4.7

Chen--Li	305--308	Kähler--Einstein	symplectic surface	mean curvature flow
----------	----------	------------------	--------------------	---------------------

$$\lambda_k^2 = |A|^2(x_k, t_k) = \max_{t \leq t_k} |A|^2, \quad x_k \rightarrow p, \quad t_k \nearrow T, \tag{8.1}$$

$$F(p, T) = X_0 = 0$$

$$F_k(x, t) = \lambda_k \Big(F(x, t_k + \lambda_k^{-2} t) - F(p, t_k) \Big). \tag{8.2}$$

8.1	q_k	$q_k \rightarrow 0$
-----	-------	---------------------

$$|H| \leq \sqrt{2} |A|$$

$$|F(p, t_k) - F(p, T)| \leq \int_{t_k}^{T'} |H(p, t)| \, dt \leq C \sqrt{T - t_k}. \tag{8.3}$$

$$\lambda_k \sqrt{T - t_k} \leq C$$

$$|q_k| := |\lambda_k F(p, t_k)| \leq C. \tag{8.4}$$

$q_k \rightarrow q$	$q = 0$	307	“we assume that
$\lambda_k F(p, t_k) \rightarrow q$	q	4.7	

8.2

$$(8.1) \text{--}(8.2)$$

$$|A_k|^2(x_k, 0) = 1, \quad F_k(x_k, 0) = \lambda_k (F(x_k, t_k) - F(p, t_k)). \tag{8.5}$$

$$p$$

$$|A_k|^2(p, 0) = \lambda_k^{-2} |A|^2(p, t_k). \tag{8.6}$$

306	$ A_k ^2(p, 0) \geq c > 0$	$x_k \rightarrow p$	$ A ^2(p, t_k) \geq c \lambda_k^2$	p	x_k
	$d(x_k, p)$	(8.6)	p	ambient ball	

$$\lambda_k|F(x_k,t_k)-F(p,t_k)|\leq C\tag{8.7}$$

$$(8.6) \qquad (8.7) \qquad \text{smooth-limit} \qquad |A_\infty|(0,0)>0$$

8.3 cutoff

$$s=t_k+\lambda_k^{-2}t \quad P_k=F(p,t_k) \qquad d\mu_t^k=\lambda_k^2d\mu_s$$

$$F_k+\lambda_kP_k=\lambda_kF, \qquad t_k-s=-\lambda_k^{-2}t.$$

cutoff

$$\int_{\Sigma_t^k} \frac{1}{v} \, \phi_R(F_k) \frac{1}{-t} \exp\bigg(-\frac{|F_k+\lambda_kP_k|^2}{4(-t)}\bigg) \, d\mu_t^k$$

$$= \int_{\Sigma_s} \frac{1}{v} \, \phi_R(\lambda_k(F-P_k)) \frac{1}{t_k-s} \exp\bigg(-\frac{|F|^2}{4(t_k-s)}\bigg) \, d\mu_s.\tag{8.8}$$

$$306 \qquad \phi_R(F) \qquad \text{cutoff}$$

$$\chi\in C_c^\infty \qquad \chi=1 \quad X_0 \qquad \lambda_kP_k \qquad \text{rescaled}$$

ball $|Y|\leq R$

$$F=P_k+\lambda_k^{-1}Y\longrightarrow X_0\tag{8.9}$$

$$(8.8) \qquad k \quad \chi(F)\equiv 1 \qquad \text{fixed original-coordinate cutoff} \qquad \phi_R(F_k)$$

k -dependent moving cutoff

8.4

monotonicity compactness smooth immersed surface $F_\infty:\Sigma_\infty\rightarrow\mathbb{R}^4$

$$H_\infty=0, \qquad (F_\infty+q)^\perp=0.\tag{8.10}$$

$$\Omega=\{F_\infty+q\neq 0\} \qquad F_\infty+q \qquad \text{tangent vector} \qquad F_\infty+q=dF_\infty(W) \qquad \text{tangent}$$

vector X

$$D_X(F_\infty+q)=dF_\infty(X)$$

$$\text{normal component} \quad \text{Gauss} \qquad A_\infty(X,W)=0 \qquad e_1=W/|W|$$

$$A_\infty(e_1,e_1)=A_\infty(e_1,e_2)=0.$$

$$H_\infty=A_\infty(e_1,e_1)+A_\infty(e_2,e_2)=0 \qquad A_\infty(e_2,e_2)=0 \qquad A_\infty=0 \quad \Omega \qquad \Omega \qquad F_\infty+q$$

$$F_\infty \qquad \text{immersion} \qquad A_\infty\equiv 0$$

$$q \qquad q=0$$

8.5

compactness monotonicity $q = 0$

Theorem 8.1 q **blow-up contradiction** $G_k : M_k \rightarrow \mathbb{R}^N$ rescaled time-slices
 $p_k \in M_k$

$$|A_k|(p_k) = 1, \quad q_k \rightarrow q \in \mathbb{R}^N, \quad \theta_k \rightarrow \theta \in (0, \infty). \tag{8.11}$$

U $p_\infty \in U$ compact exhaustion $K_i \Subset U$ K_i p_∞ embeddings
 $\Phi_{k,i} : K_i \rightarrow M_k$

$$\Phi_{k,i}(p_\infty) = p_k, \quad G_k \circ \Phi_{k,i} \longrightarrow G_\infty \quad \text{in } C^2(K_i). \tag{8.12}$$

i $\varepsilon \in \{-1, 1\}$ localized estimates

$$\int_{K_i} |H_k|^2 d\mu_k \longrightarrow 0, \tag{8.13}$$

$$\int_{K_i} \left| H_k + \frac{\varepsilon}{2\theta_k} (G_k + q_k)^\perp \right|^2 d\mu_k \longrightarrow 0, \tag{8.14}$$

contradiction q_k $q = 0$

Proof. (8.12) induced metrics area densities mean curvature vectors normal projections K_i
(8.13)--(8.14)

$$H_\infty = 0, \quad H_\infty + \frac{\varepsilon}{2\theta} (G_\infty + q)^\perp = 0 \quad \text{on } K_i^\circ. \tag{8.15}$$

K_i° U $\theta > 0$ $H_\infty = 0$ $(G_\infty + q)^\perp = 0$ U 8.4 $A_\infty \equiv 0$ (8.11)-
(8.12) C^2 convergence

$$|A_\infty|(p_\infty) = \lim_{k \rightarrow \infty} |A_k|(p_k) = 1, \tag{8.16}$$

Theorem 8.1 Chen--Li rescaling $|A_k| = 1$
pointed compactness (8.8) fixed original-coordinate cutoff
(8.13)--(8.14) p (8.7) x_k $G_k(x_k, 0) = 0$
Gaussian localized estimates x_k
 p

9. fixed cutoff

transfer proof fixed-center Proposition 5.1 ball inclusion time translation weighted
monotonicity cutoff ϕ

$$\phi(F) = \phi\big(X_0 + \lambda_j^{-1}(F_j + q_j)\big).$$

$|F_j| \leq R \quad |q_j| \leq Q \quad \lambda_j^{-1}(F_j + q_j) \rightarrow 0 \text{ uniformly} \quad j \quad \text{cutoff} \quad (3.2)$

Gaussian weight H_j, V_j unweighted estimates (2.5) f_j monotonicity square (2.11)

shifted normal-position estimate moving cutoff

10.

1.	369	curvature-normalizing moving-center scaling	Han--Li--Sun	fixed-center tangent-flow scaling
2.	“	Proposition 5.1	”	shrinking sphere ordinary MCF
3.	beta-symplectic flow			beta-symplectic
	q_j, θ_j	center-control hypotheses	fixed-center proof	moving-center version
		smooth nonflat normalized limit		
4.	q_j, θ_j	fixed-center Proposition 5.1		$q_j^\perp \quad L^2$
	$q_j \rightarrow 0$	local area bound		
5.		blow-up	fixed-center scaling	tangent behavior curvature-normalizing
		moving-center scaling		nonflat singularity model
6.	$F(x_j, t_j) \rightarrow X_0$		$q_j \rightarrow 0 \quad \lambda_j \asymp (T - t_j)^{-1/2}$	sharp
	$ F(x_j, t_j) - X_0 = o(\sqrt{T - t_j})$			
7.	Chen--Li	4.7	q_k	$q \quad q = 0 \quad x_k$
	p	cutoff		
8.		pointed smooth compactness	localized monotonicity	$H_\infty = 0$
	$(F_\infty + q)^\perp = 0$	$A_\infty = 0$		$q = 0$

1. B. Andrews, B. Chow, C. Guenther, and M. Langford, *Extrinsic Geometric Flows*, Graduate Studies in Mathematics 206, American Mathematical Society, 2020, especially printed pp. 369 and 372--374. DOI: 10.1090/gsm/206.

2. X. Han, J. Li, and J. Sun, *Gradient flow for beta-symplectic critical surfaces*, Ann. Inst. H. Poincaré Anal. Non Linéaire 41 (2024), 1083--1116, especially Proposition 5.1. DOI: 10.4171/AIHPC/100.

3. J. Chen and J. Li, *Mean curvature flow of surface in 4-manifolds*, Advances in Mathematics 163 (2001), 287--309, especially Theorem 4.7 on printed pp. 305--308. DOI: 10.1006/aima.2001.2008.